OpenSourceRail

v0.6 - Wheel-supported vehicle reference

Simple vehicles. Repeated civil works. Comfortable, frequent journeys.

Reference engineering and planned prototype work | 31 October 2026

Elevated train and station. A wheel-supported train on conventional rails beside an elevated station, with roof solar panels and a platform canopy.

Concept artwork illustrates the intended railway. Controlled geometry and engineering records govern dimensions and equipment; these images do not establish physical performance or qualification.

Six lightweight reference cars run on passive wheel bogies and conventional steel rails. The retained passive aluminium secondary is an interface reference, not a selection of installed LIM propulsion. The reference train has no traction motors, traction battery or permanent magnets. Installed propulsion is evaluated separately for each candidate.

Selected for development, with explicit trade-offs. Physical operation, lowest whole-life cost, complete installed mass and manufacturing release remain unestablished.

The whole OpenSourceRail project

OpenSourceRail develops the railway and the open tools needed to plan, design, procure, manufacture, construct, commission, operate, maintain and renew it. The aim is to retain design authority, software, fabrication, integration and repair capability in-country. Changing the reference vehicle does not remove that wider remit.

Connected product	Useful scope	Present boundary
Design and delivery platform	City inputs, candidate revisions, quantities, supplier enquiries, project and lifecycle evidence.	Research workflows and configurable business/supervision components; complete current-family deployment integration remains open.
Railway and manufacturing system	Vehicles, stations, civil works, material/process choices, tooling, assembly and repair.	Physical candidates and numerical methods; installed equipment, production data and qualification remain incomplete.
Open GoA4 controls	Station-led dispatch, ATP, interlocking, supervision and independent fault response.	Rust research, simulation and method/formal checks; no live railway or operating release.

The OSR ERPNext/Frappe HR app supports draft project, procurement, workforce, stock, manufacturing, quality and maintenance workflows. The FUXA/gateway integration supports equipment observations, trends, alarms, history and review-linked lifecycle records. Deployment needs legal entities, users, business masters, device bindings, permissions and reviewed imports.

The current engineering Workbench is read-only. The planning workspace edits route, stop, service and origin-destination proposals over current city geography and map context. Compile an exported bundle with ./osr plan compile edits.json --output build/my-plan and verify it with ./osr plan verify build/my-plan. The immutable handoff is bound to the selected v0.6 candidate and source identities. It carries proposals only; it excludes vehicle, propulsion, fleet, civil quantities, cost, performance and cashflow results, and it grants no movement authority. Missing prices may be covered in a separate low/base/high Class C estimate from calculated quantities and declared assumptions; supplier quotes are not required for design-stage comparison. Full installed city costs, candidate-specific route performance, procurement groups and construction schedules require their own evaluated route and site evidence.

A controlled first adoption can organise a workshop, depot or bounded planning pilot before a new train is physically qualified. A business case closure, cleared alarm, source hash or simulated inspection cannot accept manufactured hardware or authorise movement.

Start with README.md, docs/first-adoptable-product.md, docs/ARCHITECTURE.md and the deployment/erpnext/ and deployment/supervision/ guides for current interfaces and configuration.

Local capability, manufacture and affordability

The objective is a simple, repairable railway with low whole-life cost. Develop concrete precasting, structural fabrication, GFRP panels, winding, interiors, wiring, assembly, inspection and maintenance locally. Specialist components remain competitively sourced and must retain their actual origin.

Decision	What the comparison must include
Vehicle and equipment	Complete installed mass, passenger comfort, accessible layout, joints, thermal/fire behaviour, supplier evidence and repair access.
Civil construction	Rails/fastenings, beams, torsion, bearings, piers, foundations, drainage, containment, emergency access, transport and erection.
Manufacturing	Purchased stock, cuts, scrap, EBOM/MBOM, moulds/extrusions, tooling, joining, assembly precedence, labour, inspection and maintenance.
Service and economics	Capacity, protected headway, dwell, fleet, stations and power, construction, purchased energy, availability, maintenance and renewals.

A lighter body can reduce motor demand and civil loading, but a cheaper train can still require a more expensive station or guideway. The same candidate must carry its physical, operating and manufacturing consequences through the comparison. Unknown equipment and prices stay unknown.

Local content means origin-traced domestic value added. A distributor address does not establish where cells, semiconductors, bearings, materials or machinery were manufactured. Financing and foreign-currency requirements need supported procurement and programme assumptions, separately from physical resource cost.

Current candidate packages generate CAD/IFC stock, engineering and manufacturing records, assembly/procurement tasks and disabled ERPNext drafts. They are not submitted purchases, qualified shop instructions, an as-built operating twin or a complete priced construction/cashflow plan.

Current cost scenarios are calculated from candidate quantities, dated rates and declared assumptions. They do not establish supplier prices, savings or domestic-content percentages. All 266 city geographies remain planning inputs; candidate performance publication remains held.

Contributors can improve material/supplier evidence, complete assemblies, physical models, manufacturing and repair methods, controls, city data and tests. See CONTRIBUTING.md and GOVERNANCE.md. Software uses Apache 2.0, open physical designs CERN-OHL-S v2 and documentation CC-BY-SA 4.0, subject to the notices in LICENSE.md.

Passenger comfort and local assembly

Concept illustrations of the seated passenger experience and local manufacturing objective. Detailed equipment, accessible clearances and assembly sequences remain engineering work.

Accessible passenger interior. A bright seated cabin with a wheelchair space beside the doors. Detailed seat counts, boarding clearances and restraints require the controlled layout.

Local vehicle assembly. Workshop assembly concept showing the body, roof, glazing, doors, seating and bogies as separate assemblies.

Precast construction and station propulsion

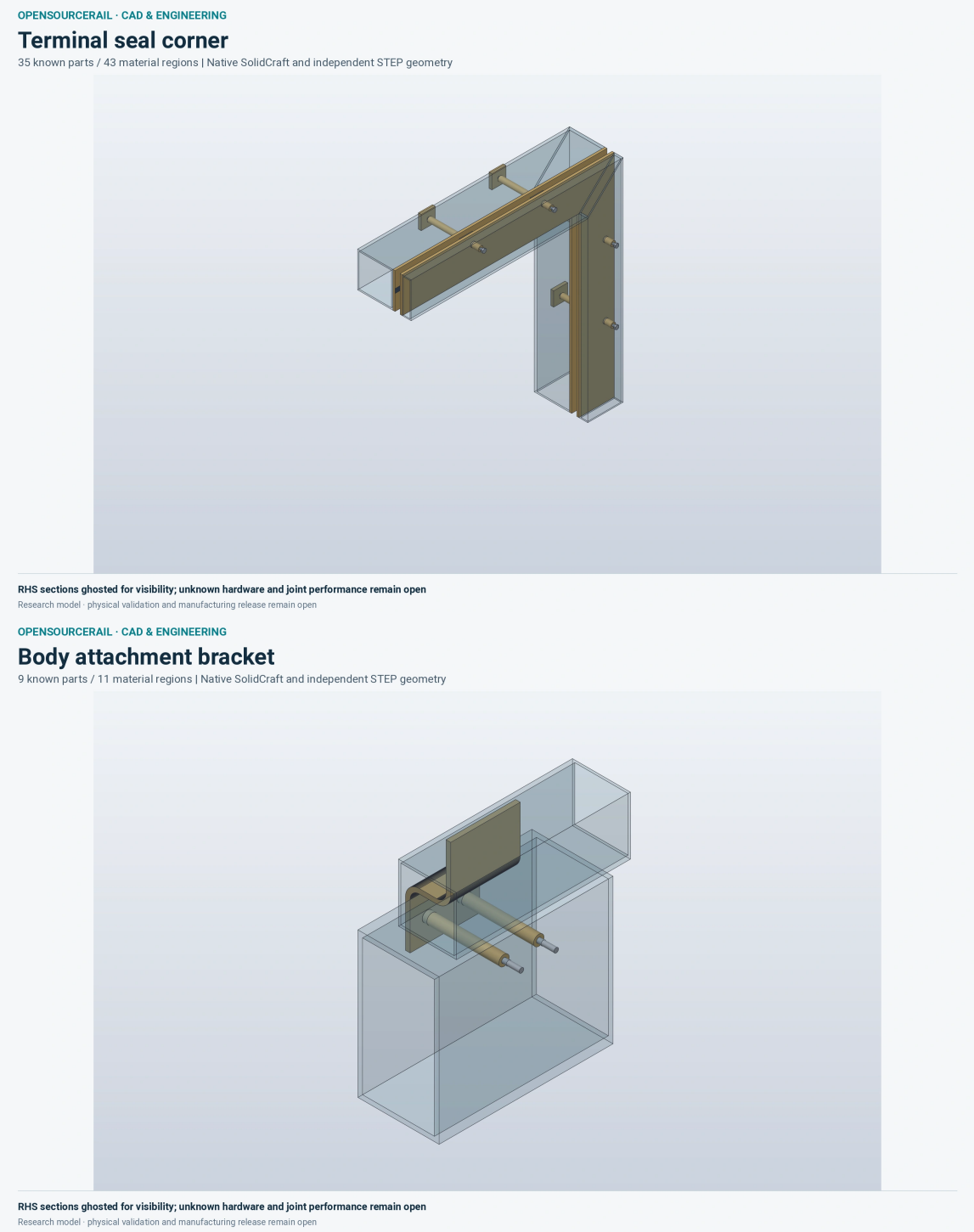
Concept illustrations connect repeated civil works with wheel-supported rail. The schematic LIM alternative is not an installed propulsion selection; motor clearances, supports and powered lengths require the candidate engineering records.

Precast guideway construction. Illustrated precast-beam erection above an urban corridor. Lifting arrangements, temporary stability and site protection need their own engineered plans.

Station LIM alternative and passive secondary. This station LIM alternative shows passive wheel bogies and station-side double-sided primaries acting on a passive aluminium fin. Blue arrows illustrate the propulsion concept; no installed propulsion architecture or powered length is selected.

Terminal joint details from native CAD

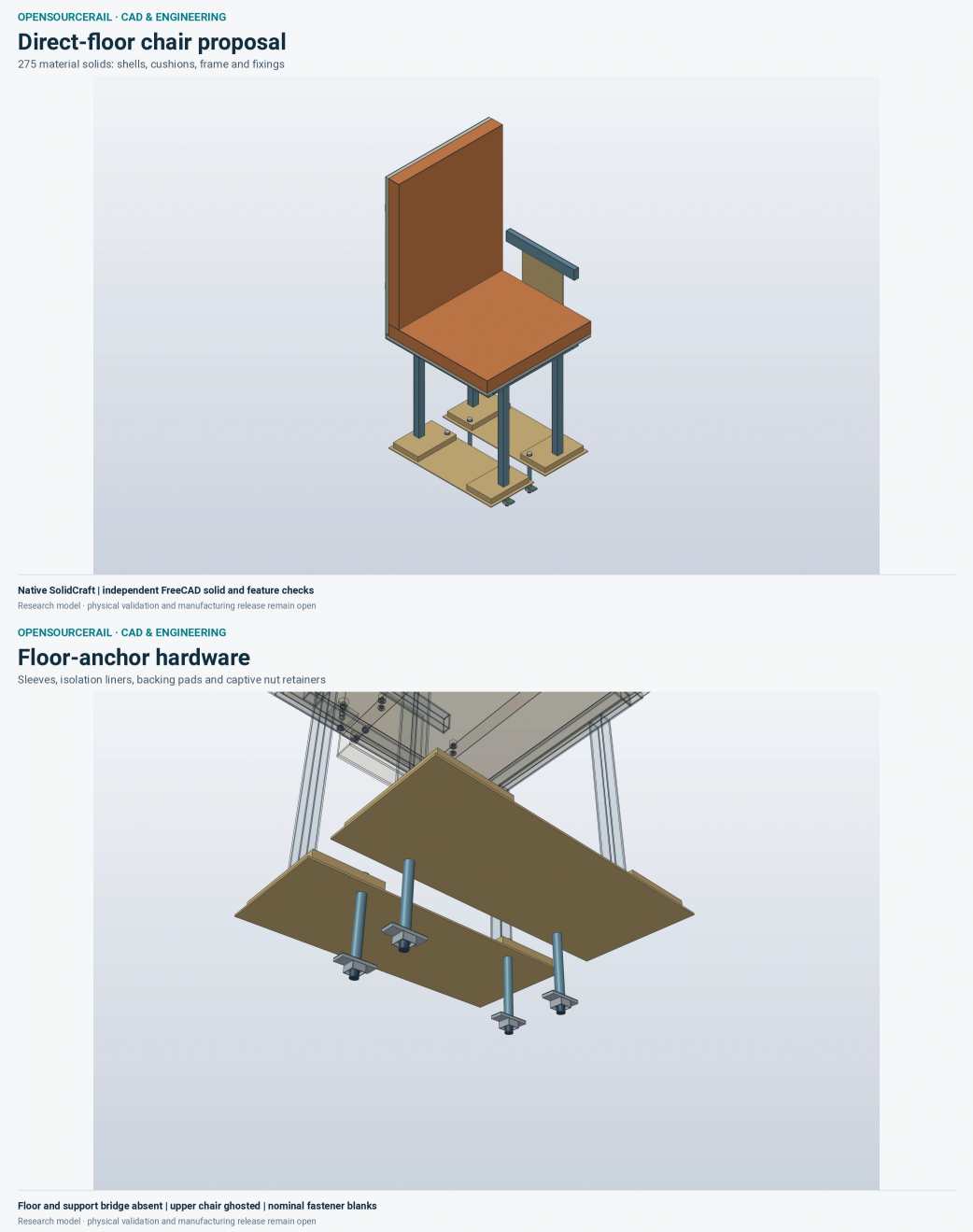
Two separate local research assemblies: a sealing-land corner and a folded body attachment bracket. SolidCraft generated and reopened both native and STEP files; FreeCAD independently checked 54 valid closed solids by individual volume, centroid and bounds. Imported part names failed the strict identity check and remain recorded as failed.



Surrounding sections are ghosted for visibility. Unknown hardware and the free gasket are excluded. Sealing response, joint strength, fatigue and installed fit remain open. Test-piece volumes are not extra installed vehicle stock. See docs/engineering-gallery.md for native files, commands, provenance and the read-only audit.

Chair and anchor hardware from native CAD

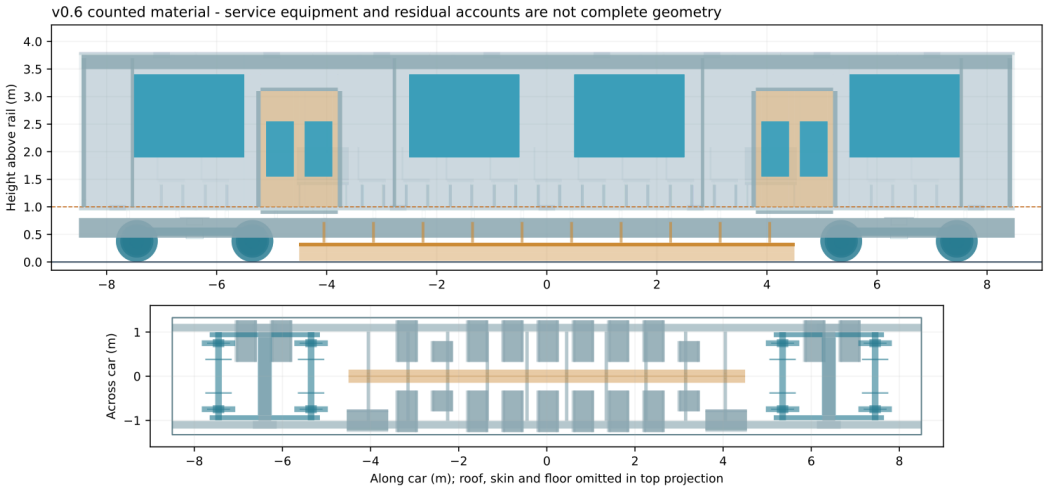
One local direct-floor chair proposal contains 275 material solids: aluminium frame and mounting plates, sandwich shells, cushions/covers, insulating liners and nominal fixing blanks. SolidCraft generated and reopened the native and STEP models. An independent FreeCAD import uniquely matches all 275 positive, valid, closed solids by volume, centroid and bounds; all 2,427 finite material/void probes agree. Imported part names failed their separate strict check and remain recorded as failed.



The underside view exposes sleeves, backing pads and captive retainers; the upper chair is ghosted. The mating structural floor and support bridge are absent. Weld, finish, seam and sealant quantities remain separately bounded. Finite probes do not establish complete geometric equivalence, interpart fit, load capacity, ergonomics or physical qualification. This coupon is not installed in the selected reference; different mounting variants need applicable evidence. Download CAD, source dimensions, material assumptions and checks from docs/engineering-gallery.md.

1. Preserve the ethos; change the support

OpenSourceRail aims to reduce whole-life transport cost through repeated components, local manufacture, accessible repair and open engineering. v0.6 removes magnetic levitation and its continuous steel-facing assemblies. Propulsion architecture, safe stopping and the cost of the complete railway still require candidate-specific evidence.



Generated counted stock geometry. Supplier reference assemblies and residual accounts are not complete installed geometry.

Location	Reference interfaces and candidate scope
Train	Six 17 m aluminium/GFRP comfort cars; identical passive bogies, independent brakes, ATP, communications, service battery, HVAC and solar roof.
Ordinary guideway	Conventional running rails and direct fastenings on precast concrete; box, twin-beam and U-beam alternatives.
Stations	Candidate-specific launch/receiving and power/charge equipment, protected berth and accessible platforms; no installed drive selected.
Control	Existing station-led GoA4 architecture with continuous onboard ATP and independent mechanical braking. No mainline switches; depot exceptions only.

Each car retains 44 fixed seats plus two wheelchair places, zero planned standing and two doors on one side. Four-abreast seating and door-zone wall benches preserve the comfort choice. The named 276-people/train, 120-second service control gives 8,280 passengers/hour/direction arithmetic; it does not select or demonstrate a protected operating headway.

The revised floor reference is 1.0 m above rail to clear conventional running gear. Platform access, one-sided doors, centre-platform compatibility, evacuation and passenger admission need physical verification. The 160-person/car crowd case remains a structural and braking sensitivity.

2. Equipped vehicle and mass accounts

Per car	Reference result	Meaning
Accounted tare	19.222 t	Materials, supplier references and unlocated accounts
46-person normal load	22.672 t	75 kg/person hypothesis
160-person crowd control	31.222 t	Not the intended normal capacity
Located material	9.132 t	Native geometry subset, not complete tare
Installed tare / CG / inertia	Unknown	Residual stock and complete equipment placement unresolved

Twelve two-axle bogies give 24 axles and 48 wheels. Motors and gearboxes are absent. Suspension, bearings, disc brakes, WSP, parking and wheel odometry remain. The 6,730 kg/car bogie reference includes partial proposed material and a transparent supplier/detail residual.

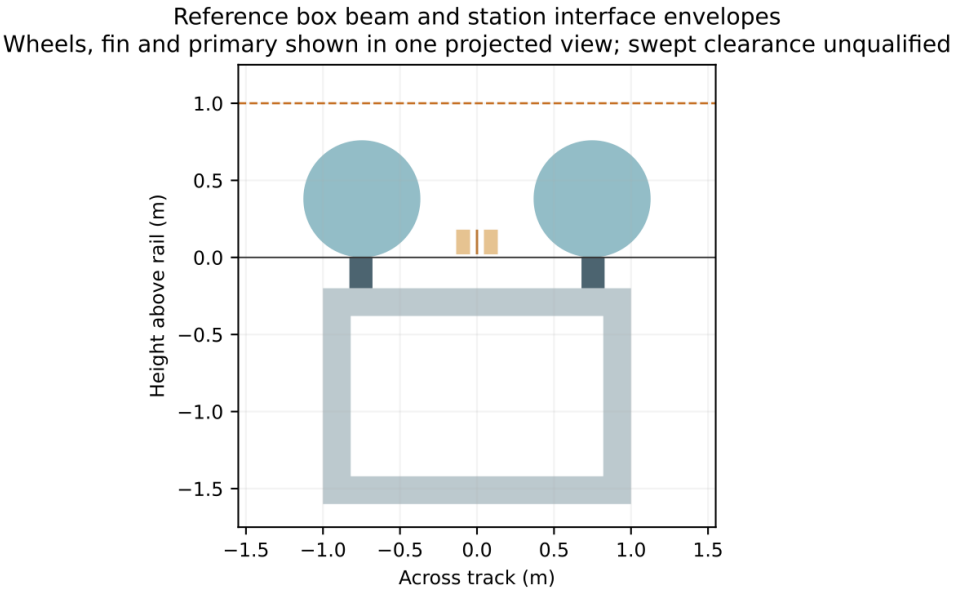
The new body has concentrated bogie reactions at 12.8 m centres rather than distributed magnetic supports. Deep side sills and the retained roof longitudinal members provide the intended body load path. The isolated side-sill fixture fails its declared stiffness limit. Ideal whole-body chord coupling is a necessary upper bound; posts, joints, shear transfer, openings, torsion, fatigue and crash behaviour remain unqualified.

Inherited body/service residuals may overlap new battery and electrical equipment. No unsupported mass subtraction is made. This conservative account needs disaggregation before it becomes a definitive equipped mass. Split battery cassettes, connections and removal paths remain unresolved.

The revised roof proposes HP60 core and a white finish. Glass, doors, floor and seating retain source ancestry; plastics remain alternatives requiring impact, fire, optics, retention and repair evidence.

3. Station motor: requirements before claims

No installed propulsion architecture or powered length is selected. The vehicle interface reference can support a separately evaluated double-sided LIM candidate. In that alternative, powered primaries and converters are wayside. Each reference car retains a 9 m passive aluminium secondary with support and heat-spreading stock. Single-sided LIM and synchronous PM alternatives remain comparisons.



Reference projected interface. Nominal 15 mm side clearance is a design target, not a demonstrated dynamic motor gap.

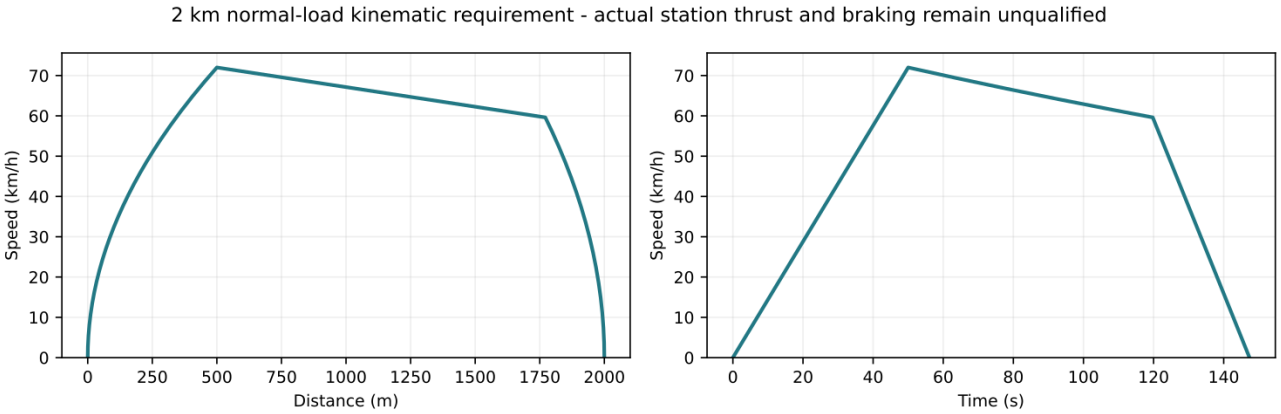
2 km normal-load requirement	Calculated target
Retained kinematic control	0.4 m/s2 /20 m/s; 500 m ideal travel only, not installed hardware
Installed powered length	Candidate-specific; no fixed selection
Sections and electrical branches	Require explicit layout and evaluated duty
Actual available thrust / voltage / current	Unknown; requires actual primary and moving-secondary field maps

The executed GetDP manufactured conductive-slab model converges against an exact imposed travelling-field solution. It verifies a numerical method. It omits the actual primary, steel, slots, finite-height current closure, motor ends and inverter circuit; conditional force is not available train thrust.

Moving suspension, wheel wear, alignment, fin expansion and unwanted motor normal forces must fit the same clearance envelope. For the LIM alternative, entry, exit, sectional handoff and repeated duty require candidate-specific evidence. The 120-second duty is a retained control, not a prescribed operating interval.

4. Reference motion and independent stopping

Input/result	Reference
Route / load	2,000 m, six normal-load cars; flat route and declared resistance
Runtime	147.362 s analytical target
Berth resource interval	Candidate-specific; no installed drive or operating headway selected
Resistance	0.0025 mg +50 N/car +1.8 v2 N/car hypothesis; not measured drag
Motion evidence	Required force, energy and geometry; physical train journey not completed



Target acceleration, natural coasting and receiving brake profile. Prescribed kinematic acceleration does not prove motor capability.

Empty, normal and crowd-control 1.5/2 km cases are evaluated, with grade and headwind sensitivities. The longest retained adjacent city gap is approximately 3 km. Adverse uphill cases can stop before receipt; downhill cases can violate the unpowered speed cap. Failures stay visible instead of adding silent boosters.

Onboard mechanical service/emergency/parking brakes must work independently of station power. A 0.05 adhesion hypothesis on a 2% downhill gives about 700 m stopping from 20 m/s including 1 s response, with no regeneration or resistance credit. Actual wet/hot curves, WSP, brake channels and thermal duty remain unknown.

5. Service energy, solar and hot weather

The auxiliary pack is for HVAC, doors, lights, communications, ATP and emergency functions only. No wheel-drive connection exists. The 82.8 kWh reference uses 75 industrial Type3-20 modules, 25 series by 3 parallel, plus enclosure, cooling, conversion, protection and retention accounts: 1,345 kg/car. This is not a railway-qualified installed pack.

Necessary energy/current check	Result and qualification boundary
35 s station charge target	200 kW/car; 1.2 MW/train before simultaneous traction and station services
Normal hotel load	35 kW/car declared reference; prototype repeated-energy bound passes
Adverse hotel load	50 kW/car requires extra charge time in the 2 km case
Hot-weather battery	Catalogue module ambient upper limit 45 C; 50 C surroundings require a proven cooled enclosure
Ageing/reserve	80% end-of-life and 90% hot-capacity hypotheses; SOC window 20%-90%
Solar	Flush roof PV retained; no unproven energy yield credited to make service pass

Catalogue voltage and continuous-current bounds do not establish charge acceptance versus SOC, temperature or life. A missed charge, stranded train and independently protected essential reserve require dispatch and emergency-energy validation. Longer/hot legs can change terminal charging, fleet and service.

LIM secondary heating is exposed over several slip assumptions. Active-plate-only and full-aluminium heat-capacity bounds differ; actual thermal conductance, repeated-duty cooling and receiving losses are unknown. A large calculated force obtained with imposed field can carry unacceptable secondary heat.

HVAC delivered cooling and COP at 50 C remain unknown. Passenger ventilation, solar gain, stationary heat rejection, redundancy, roof hot strength and durable reflective coating must close together.

6. Three civil alternatives and construction

Conventional rails replace continuous magnetic facings. Compare a closed box, two narrow beams with gauge-control diaphragms, and a U-section. Lower concrete area alone does not establish the least-cost guideway: torsion, rail support, foundations, containment, maintenance and erection matter.

Selected 20 m crowd-sized box	Counted quantity
Concrete	22.282 m3
Running rails	2.408 t
Steel incl. preliminary rebar/tendons	6.448 t
Actual installed cost	Unknown

The selected reference is 20 m with 26 prestressing strands sized to the crowd load. The same fixed beam/tendons are checked at empty, normal and crowd loads; three families across four spans retain their separate comparisons. It is a necessary longitudinal screen, not an installed-cost winner.

Moving axle-load envelopes are calculated at each longitudinal section. Each section receives its own critical train position; it does not reuse one global peak. Prestress, no-train and release stress checks and a conservative deflection envelope remain analytical screens.

The counted steel includes preliminary reinforcement and prestressing quantities. Complete piers, foundations, bearings, walkways, containment and station primaries remain outside this beam screen. Concrete and steel quantities are not installed costs, and no material reduction is converted directly to a dollar saving.

Construction sequence

Survey and geotechnical investigation; foundations; piers/headstocks; bearings and temporary stability; precast lift; rails and gauge setting; station motor installation; survey, inspection and independent release. Actual ground, crane radius, labour, factory throughput and supplier lead times are required.

Same-candidate BOM, manufacturing, procurement, assembly and controls records are research drafts. Unknown prices, availability dates and material release are not replaced with zeros or unconstrained schedules.

7. Automation, faults and city proposals

Station-led GoA4 reserves route and receiving berth, proves configuration and train readiness, sequences launch equipment and supervises receipt. Independent onboard ATP continuously enforces authority. Stations never replace the train brake protection layer.

Rust ATP/interlocking, TCMS, doors and SCADA checks exercise readiness faults and mechanical brake requests. Inputs are software fixtures: they establish neither a physical brake curve nor a safety-qualified GoA4 product.

Fault	Required independent response
Receiving drive or communications lost	Respect authority and independent stopping envelope; cut station thrust and command brakes where required
Localisation/integrity uncertain or berth occupied	Restrict authority and stop; no blind timer-based launch
Wet rail or brake channel loss	Qualified WSP and degraded brake floor; protect the larger stopping distance
Off-zone stop	No autonomous restart; protected rail-compatible rescue/towing and evacuation plan
Service power lost	Independent essential/brake supply and emergency ventilation; hot-weather endurance qualification

The 266 city geographies and detailed maps are available for planning. Calculated city outlines provide fleet, service and price scenarios; physically qualified city performance packages remain unavailable. A geography-only scan covers 13,662 adjacent-stop gaps; the largest is 7,000.0 m in Dar Es Salaam. This distance supplies an adverse analytical scenario only. Grades, sites, demand, transfers, power and prices remain city-specific.

All 266 city networks have complete calculated outlines under explicit demand, service and cost assumptions. The city dossiers expose energy, reserve-fleet consumption, capital, operations, manufacturing and finance scenarios. Local tariff and paid-hour wage estimates can replace their named accounts. Source-water civil alternatives are compared separately with obstruction reserves and interface holds; they do not silently replace headline civil designs. Physical journey and site-engineering commands retain their independent evidence holds.

8. Equal service and the selection decision

Current service basis	Development assumption
Formation	Six 17 m cars; 44 seats and two wheelchair places per car
Retained service control	120 s headway; 72 km/h launch cap; 35 s dwell; not selected operating requirements
Passenger capacity	8,280 passengers/hour/direction arithmetic; protected operation unqualified
Production and civil work	Two 8-hour shifts per working day
Operations	Two shifts for 16/18-hour service; three shifts for a 24-hour case

Neither 900 m nor 100 m is a prescribed installation. Architecture, powered length, acceleration, braking and operating headway require candidate-specific selection. Every equal-service comparison must carry vehicle, station, civil, manufacturing, staffing, energy, maintenance and renewal consequences together. Cost ranges are provisional planning estimates; supplier confirmation, equal-service route costs and full lifecycle acceptance remain open.

Release remains separate

A prototype package needs reviewed drawings, tolerances, joining, tooling, BOM, procurement, inspection and an instrumented test programme. Development adoption does not grant manufacturing, physical motion or operating release.

9. Evidence gaps and next work

Gate	State	Required evidence
installed-vehicle	unknown	Locate and weigh all residual stock, joinery, bogie and service assemblies; obtain complete CG and inertia.
isolated-underframe-stiffness	fail	The isolated hollow side-sill fixtures exceed the declared deflection limit. Whole-body load sharing through retained longitudinal members, posts and joints must be demonstrated or redesigned.
body-and-bogie-structure	unknown	Concentrated bogie support changes the body load path. Local/frame fixtures do not establish joined-body stress, buckling, fatigue, crash, modes or bogie qualification.
wheel-rail-dynamics	unknown	Qualify wheel profiles, hunting, curves, suspension, ride, flange forces and derailment containment.
station-LIM-force-map	unknown	No accepted thrust/current/frequency/gap/temperature map for the complete station drive and moving secondary; targets are required forces.
station-electrical-thermal	unknown	Qualify inverters, section handoff, supply, slip/secondary heat, unwanted normal forces and 120-second repeated duty.
independent-braking	unknown	Validate supplier brake/WSP/adhesion/thermal curves, channel failures, parking, receiving-power loss and degraded gradients.
hot-weather-services	unknown	HVAC delivered capacity and COP at 50 C, roof long-term structural temperature and actively cooled service pack remain unqualified.
coupled-journey	unknown	Reference motion is analytical. Required physical drive/brake maps, wheel dynamics and flexible-body model are not available for a qualified six-car trip.
GoA4-protection	unknown	Existing Rust ATP/interlocking and new wheel interface predicates require candidate-correlated braking, sensing, timing, communications and fault evidence.
civil-and-site	unknown	Alternative sections are necessary screens. Prestress, fatigue, torsion, lateral/derailment loads, foundations, survey, erection and independent review remain open.
rescue-and-evacuation	unknown	Motorless off-zone restart needs a protected rescue/tow system, emergency access and an acceptable high-viaduct evacuation plan.
whole-system-economics	unknown	No same-service supplier quotes or complete CAPEX/OPEX/renewal comparison; lower installed cost has not been established.
city-regeneration	hold	Preserve 266 geographic inputs. Calculated city outlines are available; physical route performance and deployment approval require candidate-specific evidence.

10. Native motor, body and brake follow-up

The bounded follow-up adds 18 current-excited GetDP motor cases and 24 actual Chrono brake executions. These are component/reduced-model experiments, not a physically coupled launch-coast-arrival journey.

Executed comparison	Result
Full 12.5 m primary and 9 m secondary	7.174 kN/car versus 11.103 kN required
Peak steel field	2.240 T exceeds declared 1.3 T linear-domain bound
Bare primary copper and steel	228.2 kg/m, excluding cooling, supports and power equipment
Full motor heat at tested point	52.1 kW secondary plus 43.6 kW primary copper per engaged module

Successful zero-current, phase-reversal and energy checks do not repair the force deficit or out-of-domain steel. Winding regrouping can reduce an arithmetic voltage requirement at much higher converter current, but finite-overlap current sharing and thermal duty are unknown. Installed available force stays unknown.

Normal-load braking from 20 m/s	5 ms Chrono result
dry	329.6 m
wet-downhill	719.0 m
one-channel-lost	1399.4 m
insufficient-adhesion	Cannot stop

Wet/downhill uses adhesion 0.05 on a 2% downgrade; lost channel retains four discs/car. Assumed pad friction, piston pressure and ideal instantaneous WSP are not qualified hardware. The crowd failed-channel case needs over 3 km. Fault detection, authority, route speed and protected overrun must use a validated degraded stopping envelope.

Independent body verification rejected native rotational beam equations and an overly rigid solid-section fixture. Rejected results remain recorded. Counted post connections and actual floor cutouts/collars add 77.154 kg/car before unknown joining hardware; they are an unadopted alternative. Native stock fit-up does not prove body strength, fatigue or complete assembly.

Counted wheel, axle and disc rotation adds 1,770.387 kg/car equivalent inertial mass, with no added static weight. It now feeds launch work, coasting and brake energy. Exact findings, benchmark dispositions and compact source-bound evidence are in docs/design-engine/v06-physical-closure/.

11. Failure-driven candidate generations

The existing runner completed 3 generations and 6 physical wheel/LIM candidates. Code generated each revision from recorded failures; the reference geometry was not manually edited between trials.

Generation /candidate	Accounted tare/car	Normal deflection	Motor validity
g0 / 190290ff88	19.299 t	111.1 mm	invalid
g0 / b96ae12040	19.306 t	32.2 mm	invalid
g1 / 035b98536f	20.231 t	27.8 mm	invalid
g1 / 7147932737	19.324 t	32.3 mm	invalid
g2 / 168c9dc7fe	20.764 t	22.7 mm	invalid
g2 / d8941f800e	19.324 t	32.3 mm	invalid

The next experiment accounts for 20.764 t/car. Its crowd deflection is 33.6 mm against 25.6 mm. Required thrust is 11.758 kN/car; the conditional calculation is 9.733 kN. Invalid motor fields cannot establish available thrust or physical feasibility.

The lighter participating-panel branch remains laterally nonconverged. Its vertical values are diagnostic. Synthetic laminate/core constants and ideal clamp transfer are unqualified; invalid body results trigger model work rather than unsupported geometry revisions.

Valid frame failures generate deeper/thicker counted stock. Motor revisions change teeth, backiron and windings; matched fin/neck geometry, car mass, rotor energy, brakes and civil loads propagate together. The fixed linear-steel limit cannot be relaxed to manufacture a pass.

Exact dependency digests, source revisions and solver identities bind caches and restart. The Pareto comparison ranks diagnostic priority and counted quantities. Prices and complete installed equipment remain unknown; research selection cannot satisfy integrated acceptance.

Every candidate generates native FreeCAD/STEP, reopened IFC, material takeoff and draft assembly, QA, procurement and maintenance records for known stock. Unknown equipment has no placeholder solids. Position-dependent journeys, joint/fatigue checks, qualified brakes and complete mass/CG/inertia remain open.

Inspect docs/design-engine/v06-optimisation/ and ./osr design campaign create/run/compare/publish. Typed next tasks retain candidate and trigger IDs with bounded scope. The 266 city packages and prior physical evidence remain preserved.

12. Challenge the railway architecture

The umbrella research campaign executes 509 analytical cases across 174 physical definitions and 2 generations. It preserves the selected v0.6 reference while testing different propulsion distribution, powered-track length, speed, formation and concrete sections.

Family	Research arrangement
P1	Current station LIM
P2	Lower-speed station LIM
P3	Sparse wayside boosters
P4	Creep/recovery drive
P5	Distributed wayside propulsion
P6	Onboard wheel traction
P7	Alternative passive secondary

Required-force trajectories compare 4/6/8 cars, 1.5/2/3 km station spacing and 43.2/57.6/72 km/h caps on a 6 km comparison ring. New drive and recovery equipment remains unweighed. Changed station density changes access service. Protected headway, actual available thrust and complete installed costs remain unknown; no economic winner or physical journey is claimed.

Six-axis finite-area joint fixtures now check work, equilibrium, reciprocity and modal refinement. The counted whole-body alternative adds 113.008 kg/car of known stock. After a retained 3 GiB allocation failure, a separately recorded 5 GiB resource revision completes the finer mesh without changing physics or limits. The vertical mesh change is 1.43%; normal deflection 28.018 mm exceeds the provisional 25.6 mm comparison. The new adaptive policy treats that comparison as advisory. Perfect bonds and ideal section transfer require physical calibration and local joint/strength/fatigue checks.

The nonlinear DC magnetic method reproduces a generic 49-point material circuit and its independent coenergy-gradient force. It is not selected motor steel or a moving nonlinear LIM. Spatial entry/centre/exit controls remain linear. Full nonlinear moving thrust, losses, inverter and thermal maps are still needed before a motor can supply a physical journey.

Capability admission binds geometry, interfaces, evaluator, benchmark, source and validity domain. Unsupported configurations generate bounded implementation or measurement tasks. M1 component correlation, M2 passive-carriage launch/coast/stop and M3 integrated pilot plans require measured data, uncertainty and independent review; none is marked complete.

Inspect docs/design-engine/v06-architecture/. Run ./osr design architecture run/publish/verify. Exact-source caching and an interrupted-run receipt demonstrate restart. The Workbench exposes all seven families, and scheduled CI runs bounded joint and nonlinear magnetic method checks.

13. Adaptive design from broad requirements

The function-first generator declares 102 extensible variables across 14 domains. Materials, honeycombs, layups, load paths, joints, support arrangements, propulsion, fitout, service and civil/construction methods can change under separately identified research configurations. Examples are seeds, not an exhaustive menu.

Mandatory functional and safety obligations remain. Quantitative hard limits need a justified interface/material/function basis. Initial dimensions, L/500-style deflection comparisons, speed and headway are explicitly revisable hypotheses or targets. Numerical-model limits stay separate: changing the old linear-steel field bound cannot create nonlinear physics.

Native section experiment	Tare account/car	Normal bending	Stock change
180 x392.187 x4.593 mm-generated	19.054 t	29.529 mm	-281.24 kg
180 x600 x4 mm-generated	19.146 t	21.272 mm	-188.70 kg

Changes are relative to the previous finite-area jointed body: 19.335 t/car and 28.018 mm normal displacement. These are counted stock plus inherited accounts, not a complete installed vehicle mass. Normal coarse/fine and crowd runs use the actual new section stock and passenger loads. Both displacement trends are checked; each crowd result has only one mesh.

The 600 mm-deep, 5 mm-wall experiment redistributes stock: about 48 kg/car less material and 19.793 mm normal bending. A 500 mm-deep alternative retains more underbody space. These model observations motivate further design work; local buckling, strength, joints, fatigue, flexible modes, swept clearances and physical correlation remain unknown.

Actual revised mass propagates to axle reactions, inertia-aware 1.5/2/3 km propulsion/energy demand and crowd-sized concrete alternatives. Available motor performance and complete installed cost remain unknown. Improving a provisional target does not establish safe operation.

The planner diagnoses physical failures, advisory targets, numerical/model problems and missing evidence separately. Repeated unsuccessful local changes escalate toward new load paths or architecture. Pending work survives invocation budgets. Exact native physics can be reused under a new policy context without changing its original receipt or pretending a new solve occurred.

Inspect docs/design-engine/v06-adaptive-design/ and ./osr design adaptive plan/run/publish/verify. The Workbench exposes the variables and native comparisons; the current reference and city geography remain unchanged. The search continues through its explicit frontier rather than declaring that one failed comparison ends the design.

14. Generate a different structure

The new structural grammar executes 5 initial recipes and 3 physical rewrites. Shared productions combine aluminium or research GFRP members, skins, foam/honeycomb sandwich layers, rib arrangements, modular segments and integrated roof functions. New combinations do not require another predefined body mesh; unsupported primitives and physics still need development.

Executed stage	Result	Evidence boundary
Generated candidates	8	Actual stock, apertures, material and joining definitions
Native CAD exports	6	FreeCAD and STEP reopen with matching stock volumes
Axial design calculations	2	Independent stiffness, equilibrium and energy checks
Complete body qualification	Unknown	Shell, joints, buckling, modes, fatigue and correlation remain open

The initial pin-jointed rectangular underfloor has 49 mechanisms. Diagnosis generates a new triangulated load path below the passenger openings, regenerates its stock and loads, and reruns it. The child has zero mechanisms; two native axial subdivisions agree with the independent matrix solution. Its approximately 30.35 mm displacement belongs to an underfloor-only hypothesis, not the previous whole-body model. No L/500 acceptance claim follows.

A separate rewrite integrates roof, solar-mount and cable/drain functions into fewer declared manufactured groups. About 44 kg of nominal stock is removed, but equivalent performance, installed hardware, manufacture, sealing and repair remain unproven. A third rewrite changes rib positions. Material search and typed interfaces hold missing hot/wet, directional, capacity and process evidence explicitly.

Gross untrimmed stock includes unresolved node overlaps and joint cuts. Known-stock CG and inertia can be calculated; complete installed mass properties and prices cannot. Source-bound equipment accounts remain unlocated. New stock propagates to the same candidate's motion demand and civil screens, not a validated force-motion-thermal journey.

A failed tripod constraint input was preserved, the writer repaired, and four fresh independent fixtures passed. Numerical model repair is distinct from physical redesign. OpenMDAO, pymoo and Dakota were not executed for this comparison. Native FreeCAD and CalculiX performed the CAD and axial work. Inspect docs/design-engine/generative-structure/ or run ./osr design generative plan/run/publish/verify.

15. One coordinated railway experiment

The new coordinator connects architecture, adaptive revision proposals, physical bodies, installed inventory, interfaces, propulsion demand, braking, civil loads, manufacture and economic coverage. It starts with five structural recipes and retains 14 body definitions after physical revisions. 112 railway bundles produce 784 linked stage records under the same six-car comfort/service comparison.

A body change assigns new physical identities and automatically invalidates dependent evaluations. New structural evidence also changes the case context even if geometry is unchanged. Exact native input, candidate, load, mesh, model source, executable and resources are checked before reuse. Stable experiment identity is separate from invocation and cache counters; pending tasks survive bounded runs.

Shared calculation	Current evidence boundary
Inventory and manufacturing	Actual stock, cuts/blanks, joints, assembly operations and source-bound quote coverage. Unlocated equipment and unknown hardware stay explicit.
Architecture and motor	Seven families and two primary-iron definitions. Changed bodies feed required force and powered-track quantities; available operating thrust is unknown.
Braking and civil	Actual accounted mass, wheel rotational inertia, nine brake hypotheses and three crowd-loaded precast alternatives. No proven protection floor or complete site design.
Economics and delivery	Scoped Pareto alternatives, matching BOM/assembly, disabled ERP research drafts and reopened IFC body stock. Installed costs and release remain unknown.

Wheel-traction and alternative-secondary configurations remove the retained passive-secondary account. Replacement drive/store/secondary equipment remains unweighed, with no zero-mass substitution. The physical-definition projection contains the actual new body, roof/material stock and selected equipment instead of stale reference-car fields.

No candidate has a fully verified installed mass, CG, inertia, energy demand and whole-life cost. Complete conditional cost scenarios exist at outline maturity. The forward journey refuses unavailable or invalid force maps. These connected research alternatives do not change the selected physical reference or qualify city performance.

Inspect docs/design-engine/coupled-railway/ or run ./osr design railway plan/run/publish/verify. GitHub Actions is enabled; the workflow requires engineering and browser contracts, with expensive native methods scheduled separately. A passing software check is distinct from physical qualification.

16. Three-dimensional bodies and transient fields

The generated metal frame and stressed-skin alternatives add counted floor/roof load paths, aperture frames, finite shared-face joints and roof services. Independent native fixtures check bending/shear refinement, intact-face transfer and orthotropic constitutive patches. Density-only or axial-only GFRP data receives no shell stiffness credit.

The first baseline grid created 152,004 cells and exceeded 5 GiB while factoring; no structural verdict followed. Aligned floor/roof profiles reduced coarse cells to about 44,000-47,000 with essentially neutral stock mass and fewer extrusion families. Original inputs, failed logs and source bytes are retained. Computational budgets do not become railway acceptance limits.

New solid model	Normal	Finer	Extra finer
aluminium frame	15.245 mm	17.283 mm	18.531 mm
aluminium stressed skin	6.035 mm	6.286 mm	Unknown

Normal/crowd incompatible-mode solid solves balance forces, moments and work/energy under intact-weld and lumped floor-load hypotheses. The frame's first refinement changed same-point deflection and energy materially, automatically triggering an unchanged-body 0.425 m experiment. Its remaining sensitivity stays open; nodal maxima are not a complete stress or whole-field convergence assessment. Torsion, through-thickness accuracy, local quadratic checks, joint/HAZ strength, fatigue, modes, thermal/fire and physical correlation remain necessary.

The new nonlinear transient magnetic kernel completed 0 of 0 bounded controls. Phase excitation, inverse H(B), net secondary current, Joule/mechanical work, stored energy and backward-Euler dissipation are explicit. Time refinement still changes thrust substantially. The short fixed conductor mask, generic steel and incomplete primary circuit/thermal/3 D model cannot supply the actual 20 m/s operating map.

Actual launch-coast-arrival runs remain not-run. Required-force profiles, synthetic forward-method tests, native component calculations, independently reviewed engineering evidence and operating qualification remain separate. The next frontier requests model/equipment data and further numerical work without inventing a feasible or cheapest railway.

17. Part-level vehicle manufacture and assembly

A separate six-car manufacturing detail proposal now contains 605 named leaf-part types in 40 assemblies. Materials, stock dimensions, quantities, individual fixings, fabrication, inspection and maintenance-removal methods are connected under one candidate identity. This does not select a new railway baseline or grant manufacturing release.

Part family	Examples and counts
Running gear and connections	12 bogies, 24 wheelsets, 48 axleboxes; brake/suspension parts, 5 intercar couplers, 5 separate gangways, bellows, frames and fixings; 2 end/rescue couplings.
Body and passenger fitout	Frames, floor, panel skins/cores, roofs, 264 chairs, 12 wheelchair places, 12 door openings/24 leaves, 110 glazing panes and positive retention.
Services and lighting	Simple replaceable LEDs, door motors/monitoring, battery racks and fittings, active cooling, HVAC/ducts, protected wiring and intercar jumpers.

The normal LED load is 216 W/car, with separate emergency/gangway lamps; declared conversion losses give about 1.55 kW for the train. The proposed 650 mm intercar gap gives 105.25 m body-plus-gap length before end projections and stopping/platform margins. These changed interfaces need their own verification.

Declared low/base/high direct parts and process allowances are USD 507,494 / 1,064,002 / 2,350,849 per six-car proposal. They exclude final assembly/commissioning, shared plant/tooling, tax, delivery and qualification. Gross stock and estimated bought masses are not verified installed mass or demonstrated savings. Inspect docs/design-engine/vehicle-parts/ or run ./osr design parts plan/build/verify.

Proposed local work uses repeatable frame cuts, controlled welding/bonding, folded 5052-H32 service enclosures, panel/core kits and accessible retained joints. Bought wheels, axles, bearings, brake equipment, glazing, elastomers and electronics still need traceable grades, supplier/process evidence and application qualification. Missing quotations do not prevent the outline comparison.

Assembly stage	Inspect before closing
Frame, floor and roof	Datum jig, bogie/coupler hardpoints, weld/bond preparation, retained service openings and roof mount layers.
Running gear and services	Wheelset/axlebox interfaces, brake/air circuits, electrical isolation, active battery cooling and supported removal routes.
Doors, panels and fitout	Door motor/reducer alignment, 24 separate edge sensor inserts, glazing retention, chairs, wheelchair access and light branches.
Intercar assembly	Five independent load-bearing couplers plus passenger bridges/bellows, positive fixings, hoses, jumpers and essential lamps.

The manufacturing catalogue: stock and interfaces

local fabricate

Full-length lower side sill

Car underframe and body-side bolster structure · 2 per assembly × 6 assemblies = 12 per train

Base unit mass	Base direct cost each
378.88 kg	**$2,700**

Unit mass low / base / high	Direct unit cost low / base / high
378.88 / 378.88 / 378.88 kg	**$1,600 / $2,700 / $5,200**

Exact declared gross blank volume × proposed density; holes, machining loss, weld buildup and installed trim are not silently deducted. Not measured net installed mass.

180 mm × 350 mm · wall 8 mm

Cut length 17,000 mm

Stock sketch · not to scale · released holes/joints and installed geometry pending

Material

Actual unit record in the source-built catalogue. The shown member is gross hollow stock, 180 x350 x8 mm and 17 m long; hole patterns, end details, joining and installed acceptance remain unresolved. The material/quantity record is not an assembled structural model.

Use the Vehicle parts link in the Workbench to search all 605 named types, select one of 40 assemblies and inspect materials, stock, manufacture, fixings, proposed tolerances, inspection and removal. Each item carries the same proposal candidate and bundle identity. Business drafts remain disabled; no purchasing, ERP or train-control transaction is performed.

Part-level outline completion, detailed installed CAD/fit, prototype validation and manufacturing/operating release are separate statuses. The current city, platform, fleet and physical engineering results do not transfer automatically to this new detail proposal.

18. Reproduce and inspect

Use ./osr status, ./osr readiness, ./osr build, ./osr verify, ./osr test, ./osr book --check and ./osr workbench. ./osr sim baghdad reports its deliberate city-evaluation hold. Compile and verify planning handoffs with ./osr plan compile and ./osr plan verify.

The broader design-space comparison adds supplier-bound PET/PVC and directional honeycomb cores, aluminium/GFRP load paths, joints/inserts, motor topologies, glazing and HVAC leads. Run ./osr design space build/verify. Necessary bounds and component fixtures guide the next experiments; they do not establish a complete train or silently change the default.

Repository records

engineering/development-baseline.toml

engineering/station_rail/brief.toml

docs/design-engine/current/README.md

docs/design-engine/current/evaluation.json.gz

docs/design-engine/current/manifest.json

docs/design-engine/current/exports/manifest.json

docs/design-engine/coupled-railway/delivery-index.json

Physical and evaluation identities

selection id

e56fd43277b64a1c40a30bc2f6c04b74 498263b6e3945a6889b01af6f4394803

candidate id

756a68870162f7559b052b05c9b20172 8c9f448e3bb61114f287848c02f2d3a3

body candidate id

9489e4f2666fd3b3c73232c86df3f12a df42a5531eb27373834cf3fecb1a3117

support component id

3e9aef846e6ec55bac01351216339461 8b37cf8c74b1100d6f8e5450ae5479d7

propulsion component id

67e2bfc4ef429a6c49e6d07095bc298c 4eb76b8386e4ee316c440ff128c30fa9

Source receipts bind the exact files used. Native method convergence, analytical screening, independent review and authenticated physical acceptance remain separate levels of evidence.